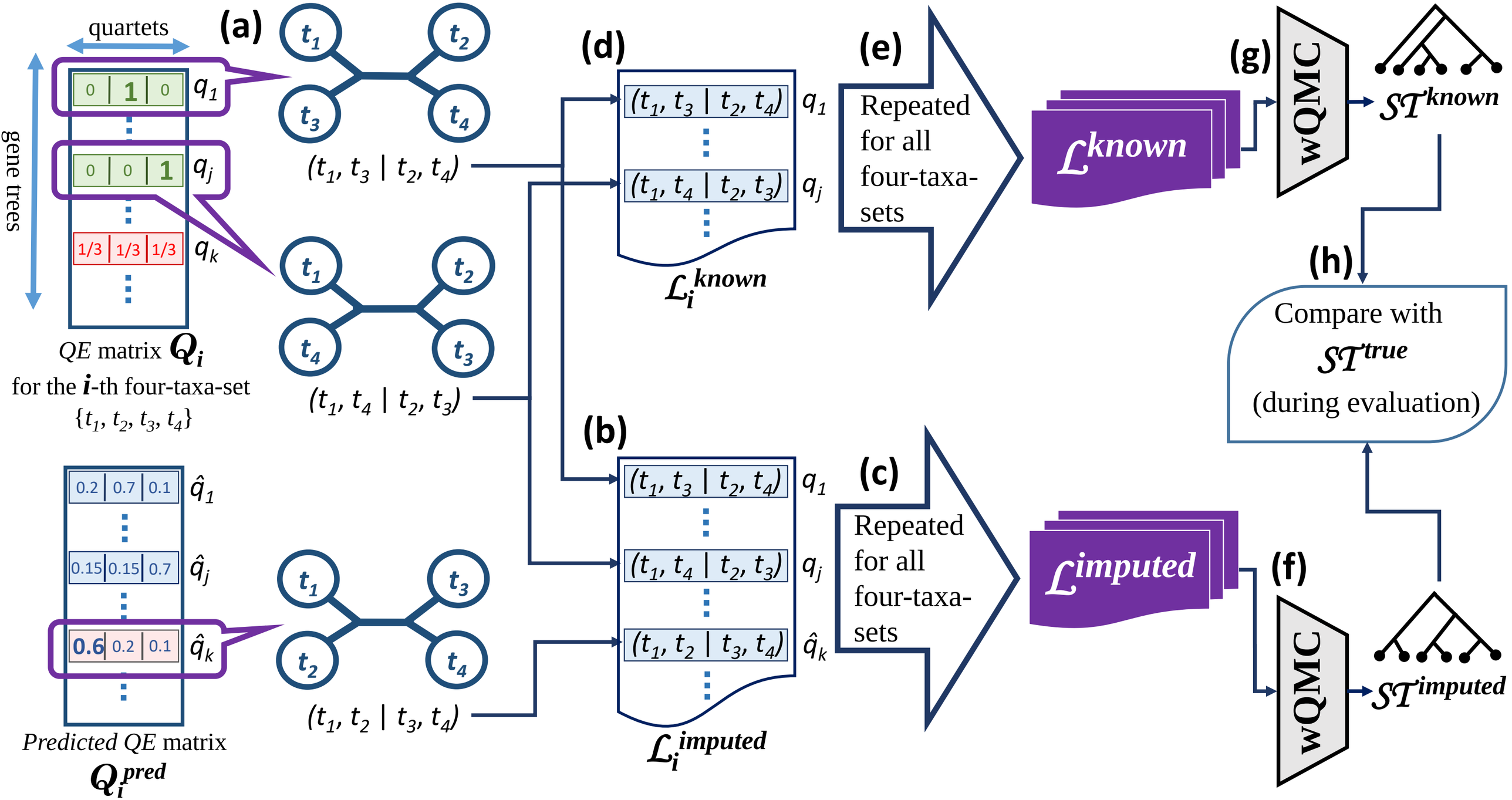

quartets
gene trees
(a)
0 | 1 | 0
q_1
0 | 0 | 1
q_j
1/3 | 1/3 | 1/3
q_k
QE matrix Q_i
for the i-th four-taxa-set
$\{t_1, t_2, t_3, t_4\}$
t_1 t_2 t_3 t_4
$(t_1, t_3 \mid t_2, t_4)$
$(t_1, t_4 \mid t_2, t_3)$
$(t_1, t_2 \mid t_3, t_4)$
0.2 | 0.7 | 0.1
$\hat{q}_1$
0.15 | 0.15 | 0.7
$\hat{q}_j$
0.6 | 0.2 | 0.1
$\hat{q}_k$
Predicted QE matrix
Q_i^{pred}
(d)
$(t_1, t_3 \mid t_2, t_4)$ q_1
$(t_1, t_4 \mid t_2, t_3)$ q_j
$\mathcal{L}_i^{known}$
(b)
$(t_1, t_3 \mid t_2, t_4)$ q_1
$(t_1, t_4 \mid t_2, t_3)$ q_j
$(t_1, t_2 \mid t_3, t_4)$ $\hat{q}_k$
$\mathcal{L}_i^{imputed}$
(e)
Repeated for all four-taxa-sets
(c)
Repeated for all four-taxa-sets
$\mathcal{L}^{known}$
$\mathcal{L}^{imputed}$
(g)
wQMC
$\mathcal{ST}^{known}$
(h)
Compare with
$\mathcal{ST}^{true}$
(during evaluation)
(f)
wQMC
$\mathcal{ST}^{imputed}$